SOC Detection Engineering Lab Report

Detection Engineering using Splunk Enterprise & Sysmon

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1. Introduction

This project documents the implementation of a Windows Detection Engineering Lab using Splunk Enterprise and Sysmon. The lab collects Windows endpoint telemetry, develops custom detections using Splunk Search Processing Language (SPL), visualizes security events through dashboards, and configures alerts for monitoring suspicious activity.

2. Objectives

- Install and configure Splunk Enterprise.
- Deploy Sysmon for Windows endpoint telemetry.
- Ingest Windows Event Logs into Splunk.
- Develop detection rules using SPL.
- Create a SOC dashboard.
- Configure scheduled alerts.
- Validate detections using simulated Windows activity.

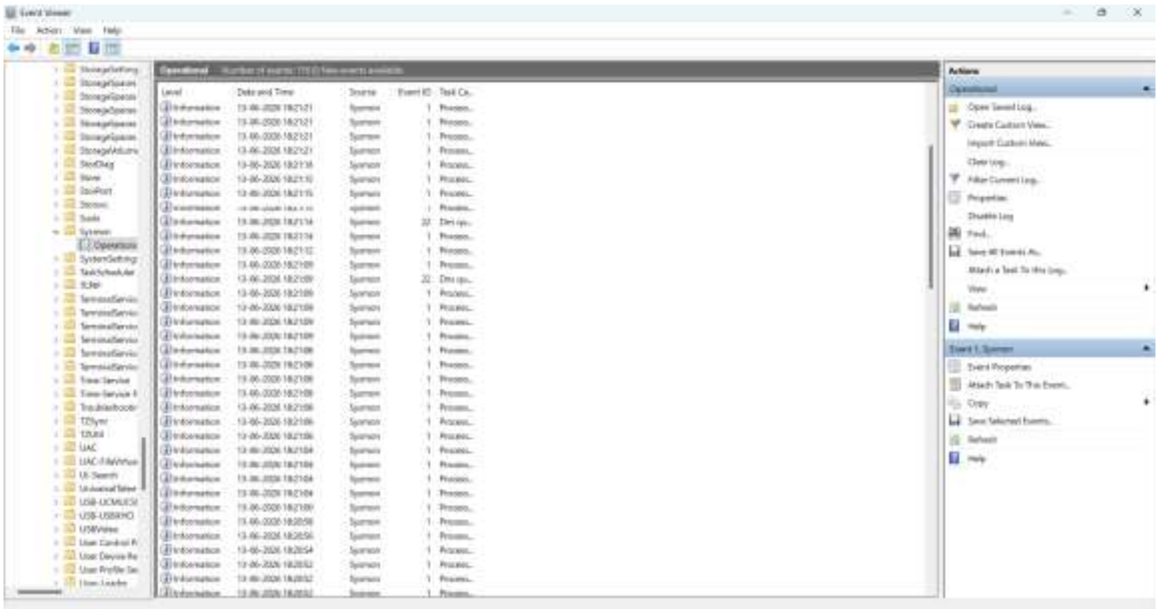
3. Lab Environment

Component	Details
Operating System	Windows 11
SIEM Platform	Splunk Enterprise 10.4
Endpoint Telemetry	Sysmon
Log Source	Windows Event Logs
Testing Tools	PowerShell, Command Prompt

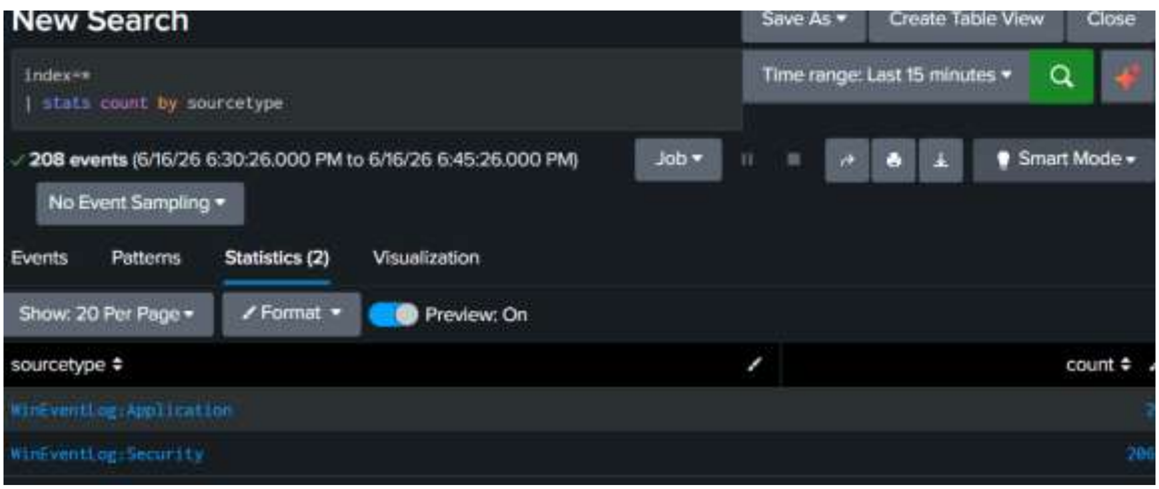
4. Project Implementation

Splunk Enterprise was installed and configured as the SIEM platform. Sysmon was deployed to generate enhanced endpoint telemetry. Windows Event Logs were ingested into Splunk and validated before creating detection rules. Test activities were generated using PowerShell, Command Prompt and Certutil.

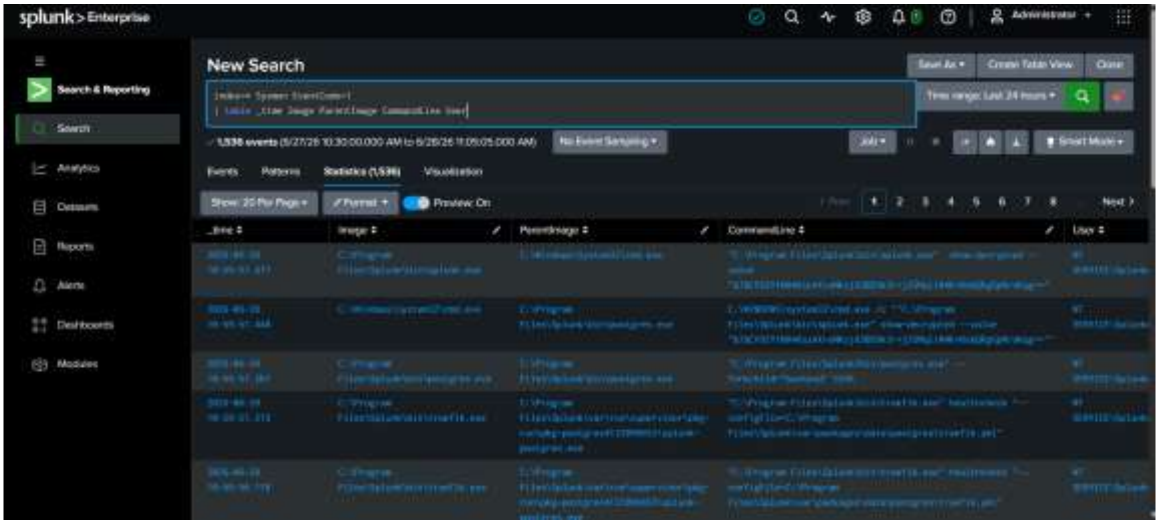
Screenshot: 01-sysmon-events-generated



Screenshot: 02-log-ingestion-validation



Screenshot: 03-process-creation-events

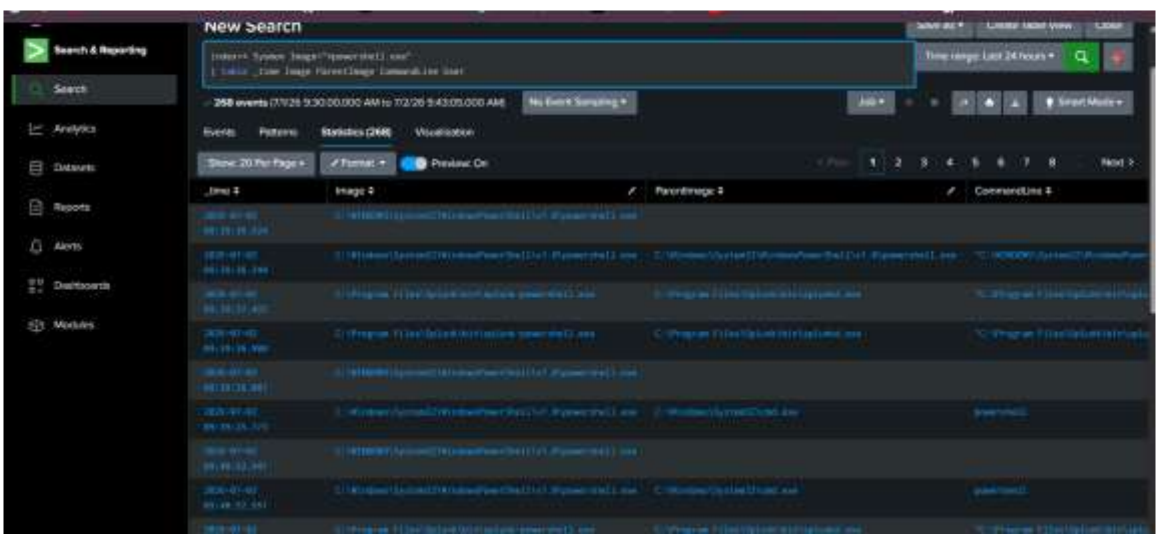


5. Detection Rules

PowerShell Execution Detection

Developed an SPL query to detect execution of powershell.exe and review the parent process, image path and command line.

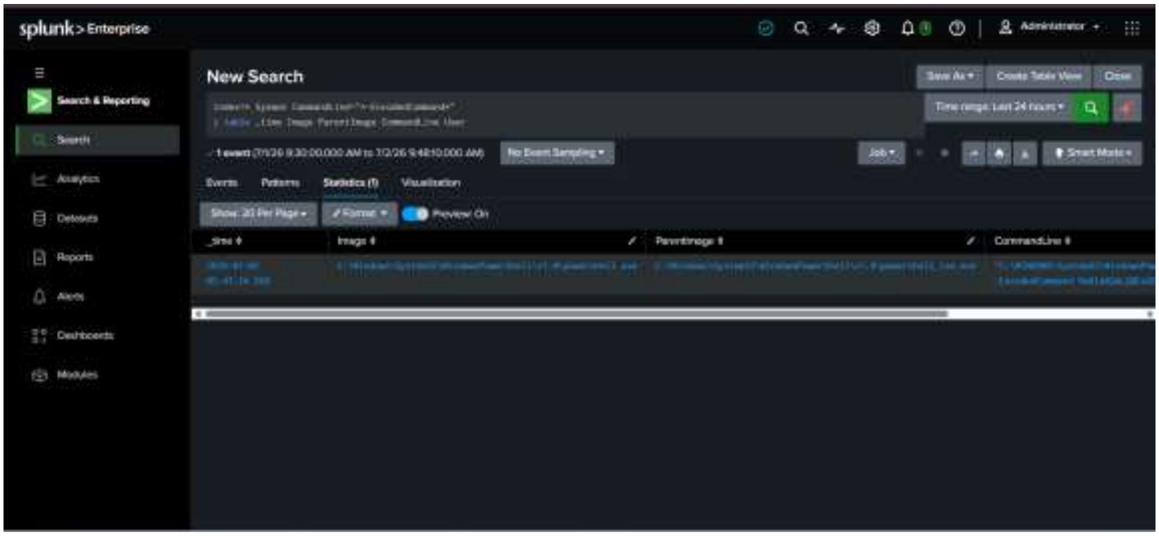
Screenshot: 04-powershell-execution-detection



Encoded PowerShell Detection

Implemented a detection for the -EncodedCommand parameter to identify obfuscated PowerShell execution.

Screenshot: 05-encoded-powershell-detection



Suspicious Parent-Child Process Detection

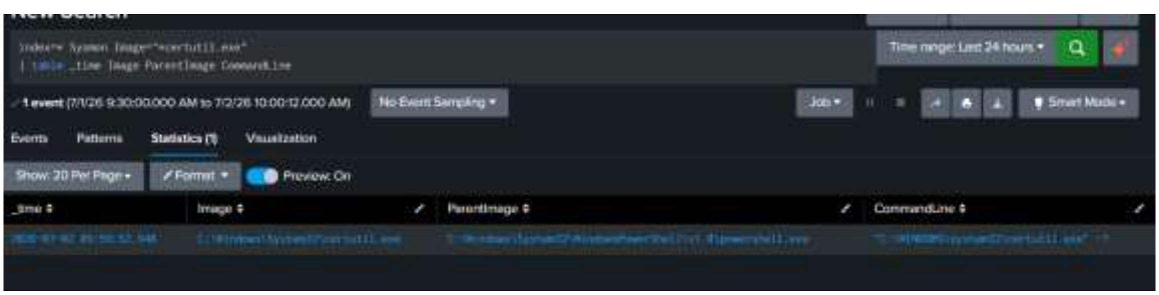
Created a detection for cmd.exe launching powershell.exe to identify suspicious execution chains.

Screenshot: 06-suspicious-parent-child-process

LOLBin (Certutil) Detection

Monitored execution of certutil.exe, a legitimate Windows utility frequently abused by attackers.

Screenshot: 07-lolbin-certutil-detection

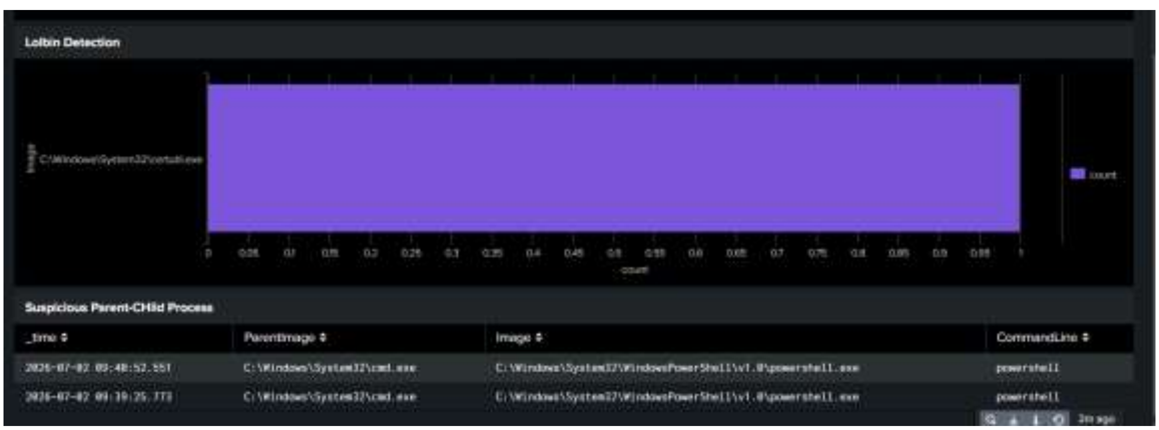


6. Dashboard

A custom dashboard was created to visualize PowerShell execution, encoded PowerShell activity, LOLBin execution and suspicious parent-child process relationships.

Screenshot: 08a-dashboard-overview

Screenshot: 08b-dashboard-overview



7. Alert Configuration

A scheduled Splunk alert was configured to detect PowerShell execution.

Screenshot: 09-powershell-execution-alert

8. Skills Demonstrated

- Detection Engineering
- Splunk Enterprise
- Sysmon Configuration
- SIEM Monitoring
- Windows Event Analysis
- Threat Hunting
- Dashboard Development
- Alert Configuration
- Splunk SPL

9. Conclusion

This project demonstrates practical experience with configuring Splunk Enterprise, integrating Sysmon, creating endpoint detections, validating alerts and investigating Windows activity. The completed lab strengthened skills relevant to Security Operations Center (SOC) operations and detection engineering.